



MONITORING PLAN

SWECC004 – Oak Tree Health Leaves

AVANS HOGESCHOOL
ESSET, Year 3 Quarter 3-4

20-06-2026

Nālani Balcells, Tibbe den
Hoed, and Bibi Versluis

Colophon

Title page, Colophon, Table of contents	Nālani Balcells
Introduction	-
Problem statement	Nālani Balcells
Project goal	Nālani Balcells
Reading guide	Nālani Balcells
General	-
Baseline conditions	Nālani Balcells
Goals for evaluation & monitoring	Nālani Balcells
Variables to be monitored	Tibbe den Hoed
Monitoring area	Bibi Versluis
Habitats & species to be monitored	Bibi Versluis
Timeframe	Bibi Versluis
Protocols/ methods to be applied	-
Frequency and responsibilities	Nālani Balcells
General oak leaf sampling protocol	Nālani Balcells
Field data recording templates	Nālani Balcells
Visual assessment protocol	Bibi Versluis
Leaf monitoring protocol	Tibbe den Hoed
Laboratory standard operating procedures	Tibbe den Hoed
Quality assurance and quality control documentation	Bibi Versluis
Supplementary Results and Supporting Data	Bibi Versluis
Outputs	-
Outputs to be produced & target users	Tibbe den Hoed
Bibliography	Tibbe den Hoed
Appendices	Nālani Balcells

Table of Contents

Colophon.....	1
1.0 Introduction	4
1.1 Problem statement	4
1.2 Project goal	4
1.4 Reading guide	4
2.0 General	6
2.1 Baseline conditions	6
2.2 Goals for evaluation & monitoring.....	6
2.3 Variables to be monitored	8
2.3.1 Crown condition indicators	8
2.3.2 Leaf-based physiological and nutritional indicators	8
2.3.3 Spatial and contextual data.....	9
2.4 Monitoring area	9
2.5 Habitats & species to be monitored	10
2.6 Timeframe	11
3.0 Protocols/ methods to be applied.....	13
3.1 Frequency and responsibilities.....	13
3.2 General oak leaf sampling protocol	14
3.3 Field data recording templates	16
3.4 Visual assessment protocol	19
3.5 Leaf Monitoring Protocol	23
3.6 Laboratory standard operating procedures	24
3.6.1 Sample preparation [5], [14].....	24
3.6.2 Acid digestion for MP-AES analysis [15], [16], [17]	26
3.6.3 Preparation of Samples [18], [19], [20]	27
3.6.4 Preparation of Calibration Standards [21], [22], [23]	29
3.6.5 MP-AES analysis [24], [25], [26]	32
3.6.6 Chlorophyll Extraction and Spectrophotometry [27], [28], [29]	33
3.6.7 Hach Lange tests protocols [23], [30].....	34
3.7 Quality assurance and quality control documentation	36
3.7.1 Field sampling.....	36
3.7.2 Laboratory analysis	37
3.7.4 Data handling and management.....	40

3.8 Supporting Reference Data	41
3.9 Tree canopy sampling - climbing and safety protocol	42
3.9.1 Scope and applicability.....	42
3.9.2 Equipment	43
3.9.3 Personnel and roles	44
3.9.4 Personal protective equipment: gloves and insect/tick safety	44
3.9.5 Step-by-step field protocol.....	45
3.9.6 Emergency procedures	47
<i>Bibliography.....</i>	48
<i>Appendices</i>	53
<i>Appendix 1 – MP-AES foliar nutrient concentrations.....</i>	53
<i>Appendix 2 – PhotoFolia Leaf Greenness Values</i>	54
<i>Appendix 3 – Visual crown assessment scores.....</i>	55
<i>Appendix 4 – Hach Lange LCK 238 & 348 results (N and P).....</i>	56
<i>Appendix 5 – Calculated nutrient ratios.....</i>	57

1.0 Introduction

1.1 Problem statement

Oak tree health at Natuurbegraafplaats Huis ter Heide has been declining, with visible signs including reduced leaf density, crown dieback, and fungal infections. Since 2022, bokashi has been applied to the soil of select trees as a remediation measure, but its effectiveness has not been systematically evaluated. The client currently lacks scientific evidence on whether bokashi improves soil health and whether this translates to improved oak vitality.

1.2 Project goal

The goal of this 20-week project is to develop, implement, and evaluate an integrated monitoring plan for oak tree (*Quercus robur*) health at Natuurbegraafplaats Huis ter Heide, by designing a coherent protocol for combined soil, leaf, and remote-sensing indicators, carrying out at least one full monitoring campaign using this protocol, and delivering a validated, practically applicable monitoring framework with a Story Map presenting the results, and an interactive GIS map that enables stakeholders to track changes in oak vitality and the effects of soil-life improving measures such as the application of bokashi, a fermentation-based soil improvement that enriches microbial activity and nutrient availability to promote soil and tree health, over time.

1.4 Reading guide

This monitoring plan is organised in three main parts, followed by a bibliography and appendices.

Section 1 (Introduction) introduces the problem context, the project goal, and this reading guide. It explains why systematic monitoring of the oak trees at Natuurbegraafplaats Huis ter Heide is necessary and what the monitoring plan aims to achieve.

Section 2 (General) provides the framework within which the monitoring takes place. It describes the baseline conditions of the study site, sets out the specific evaluation and monitoring goals, defines the variables that will be measured, outlines the monitoring area, describes the habitats and species covered, and presents the planned timeframe for the 2026 season and beyond.

Section 3 (Protocols and methods) contains all operational guidance needed to carry out the monitoring plan. It covers the recommended monitoring schedule and responsibilities (Section 3.1), the general oak leaf sampling protocol (Section 3.2), field data recording templates (Section 3.3), the visual crown assessment protocol (Section 3.4), the leaf

monitoring protocol (Section 3.5), laboratory standard operating procedures for sample preparation, acid digestion, and ICP-AES analysis (Section 3.6), and quality assurance and quality control documentation (Section 3.7).

The Bibliography lists all sources cited in this document. Supporting materials, including supplementary data tables and figures, are provided in the appendices.

2.0 General

2.1 Baseline conditions

Natuurbegraafplaats Huis ter Heide, a natural burial site located on the edge of the Loonse en Drunense Duinen, a Dutch national park in the province of North Brabant. The site covers 35 hectares of the total 650-hectare nature reserve; this is a glacial ridge characterised by well-drained, acidic, and nutrient-poor sandy soils (podzols). The surrounding landscape is predominantly forested, with Scots pine (*Pinus sylvestris*) and pedunculate oak (*Quercus robur*) as the dominant tree species. The burial ground is managed with the explicit aim of maintaining and promoting biodiversity alongside its morgue function [1].

Before the initiation of this monitoring plan, visible signs of declining health had been recorded in a number of the oak trees at the site. Observed symptoms include reduced leaf density in the upper crown, crown dieback (having progressive death of the peripheral branches and fine branch tips), epicormic growth along the trunk and main branches, and the presence of fungal infections. These symptoms are consistent with oak decline, which typically results from a combination of primary stressors, including soil compaction, nutrient limitation, periodic drought stress, and secondary agents such as pathogenic fungi and defoliating insects.

In response to these observations, the site manager began applying bokashi (a fermented organic soil amendment designed to encourage microbial activity and improve nutrient cycling and availability) to the soil surrounding a subset of trees from 2022 onwards. Three of the seven permanently monitored trees (T01, T02, T03) have received bi-annual bokashi applications; the remaining four (T04, T05, T06, T07) serve as untreated reference trees and have received no soil corrections. No assessments of tree vitality were carried out before or during the bokashi applications, which means the effect of the intervention has not yet been scientifically evaluated.

This monitoring plan establishes the first quantitative baseline against which change in oak vitality can be assessed. The initial measurements collected during Round 1 of the 2026 season (early to mid-May 2026) will serve as the baseline dataset for all subsequent comparisons, both within the 2026 season and across future growing seasons.

2.2 Goals for evaluation & monitoring

The overarching goal of this monitoring plan is to provide the site management of Natuurbegraafplaats Huis ter Heide with a scientifically grounded, practically applicable

framework for tracking oak tree vitality and evaluating the long-term effects of bokashi soil amendment. The specific evaluation and monitoring goals are as follows.

1. Establish a quantitative baseline

To record quantitative values for a defined set of oak vitality indicators; including crown condition scores, leaf nutrient concentrations, and foliar chlorophyll content for all seven monitored *Quercus robur* individuals at the start of the 2026 growing season. These values will serve as the reference point against which future measurements are compared.

2. Detect differences between treated and reference trees

To assess whether bokashi-treated trees (FT01–FT03) show measurable differences in vitality indicators relative to untreated reference trees (NFT04–NFT06, HT07), thereby providing evidence for or against the effectiveness of the intervention after at least three years of treatment.

3. Track within-season variation

To document changes in oak vitality indicators across a minimum of three phenologically distinct monitoring rounds per growing season, enabling the identification of seasonal patterns and the detection of stress that may be temporary or phenology dependent.

4. Produce spatially referenced monitoring data

To link all field measurements to georeferenced tree locations, enabling integration with drone-derived multispectral imagery (NDVI) and visualisation in a GIS environment and Story Map, so that patterns in vitality can be communicated to site managers and other stakeholders.

5. Deliver a transferable monitoring protocol

To produce validated and standardised monitoring protocols that can be independently implemented by site management staff in subsequent growing seasons without requiring specialist ecological expertise for field measurements, ensuring continuity of the long-term monitoring beyond the duration of this project.

2.3 Variables to be monitored

The monitoring plan tracks three categories of variables: crown condition indicators, leaf-based physiological and nutritional indicators, as well as spatial and contextual data. Together these provide a multi-dimensional picture of oak vitality.

2.3.1 Crown condition indicators

Crown condition is assessed visually during each monitoring round using standardised methods based on the ICP Forests protocol [2] . The following variables are recorded for each tree.

- **Crown defoliation:** the estimated percentage reduction in leaf density relative to a fully foliated reference crown of the same species, scored in five classes (Class 0: 0–10%; Class 1: >10–25%; Class 2: >25–60%; Class 3: >60–99%; Class 4: 100% dead).
- **Crown transparency:** the estimated proportion of visible sky through the foliated crown zone, recorded in 5% increments. Bare branches are treated as transparent in the assessment.
- **Crown dieback:** the proportion of recently dead terminal branches and fine branch tips in the upper and outer crown, recorded in 5% increments. Dead branches in the lower crown are excluded.
- **Epicormic shoot production:** the presence and abundance of stress-induced shoots arising from latent buds on the trunk and main branches, scored on a three-point scale (0 = none; 1 = scattered, not dominant; 2 = abundant, clearly dominating trunk and lower crown).
- **Additional damage observations:** qualitative records of any fungal fruiting bodies, bark cankers or lesions, mechanical damage, insect feeding damage, leaf discolouration, and acorn production level (absent / scarce / common / abundant).

2.3.2 Leaf-based physiological and nutritional indicators

Leaf samples collected during each monitoring round are submitted to a contracted laboratory for elemental analysis by inductively coupled plasma atomic emission spectrometry (ICP-AES) following acid digestion. The elements quantified are: nitrogen (N), phosphorus (P), potassium (K), calcium (Ca), magnesium (Mg), iron (Fe), manganese (Mn),

copper (Cu), and zinc (Zn). All concentrations are expressed on a dry-weight basis (g kg^{-1} for macronutrients; mg kg^{-1} for micronutrients) and interpreted against published sufficiency ranges for *Quercus robur*. The nutrient ratios N:P and N:K are calculated alongside absolute concentrations, as stoichiometric imbalances can indicate growth limitation even when individual element concentrations appear adequate.

Foliar chlorophyll content is estimated non-destructively in the field using the PhotoFolia smartphone application (Colourworker) [3]. The app derives a dimensionless Leaf Greenness Value (LGV) from photographs of individual leaves taken against a calibrated colour-standard card [4]. Twelve readings are taken per tree per monitoring round, distributed across four compass positions (N, S, E, W) around the accessible crown, and the per-tree mean LGV is used as the analytical unit.

2.3.3 Spatial and contextual data

Each monitored tree is assigned a unique identifier (T01–T06) and a fixed geographic location recorded by GPS (minimum positional accuracy ± 3 m). Additional tree-level attributes recorded at initial tagging and updated as needed include diameter at breast height (DBH, cm), estimated crown height class (1–5), and bokashi treatment history. Where drone-acquired multispectral imagery is available, canopy NDVI values are extracted per tree from georeferenced rasters and linked to the field dataset, enabling spatial analysis of crown greenness at the stand scale.

2.4 Monitoring area

The monitoring area is Natuurbegraafplaats Huis ter Heide, a natural burial site located on the edge of the Loonse en Drunense Duinen, a Dutch national park in the province of North Brabant. The site covers 35 hectares of the total 650-hectare nature reserve, which is owned by Natuurmonumenten and has been managed as a natural burial ground by Natuurbegraven Nederland since October 2021 [1]. This is a glacial ridge characterised by well-drained, acidic, and nutrient-poor sandy soils (podzols). Open, wind-formed sandy terrain, heathland, fens, and mixed woodland characterise the landscape. The wider landscape is predominantly forested, with Scots pine (*Pinus sylvestris*) and mixed deciduous woodland, with pedunculate oak (*Quercus robur*) occurring throughout the forested areas of the ridge. The oak trees were largely planted between 1750 and 1850 on the sandy soil of the area [1].

Monitoring is restricted to a fixed set of seven *Quercus robur* individuals within the 35-hectare managed area. The plot boundaries are fixed for the duration of the monitoring plan; no trees are to be added to or removed from the monitored set without a formal update to this protocol document. Three trees have received bi-annual bokashi soil applications since 2022, three serve as untreated controls, and one tree with a visibly healthy crown functions as a reference tree. All seven trees are permanently marked with a numbered tag (T01–T07), georeferenced using a GPS-enabled device, and their locations are documented in the Tree Inventory Sheet (Section 3.3). Tree locations, diameter at breast height (DBH), and crown height class are recorded in the Tree Inventory Sheet (Section 3.3) and kept up to date throughout the monitoring [5].

The seven monitored trees are located within walking distance of each other and of the main site entrance, making the plot accessible on foot without specialist equipment. Of the seven trees, three (T01, T02, T03) have received bi-annual bokashi soil applications since 2022 and constitute the treatment group; the remaining four (T04, T05, T06, T07) have received no soil amendment and serve as the untreated reference group. This spatial separation of treatment and reference trees is shown in the GIS map produced as part of connected project.

Monitoring activities do not extend into the broader Natuurmonumenten reserve. Access scheduling is coordinated by the site manager to avoid interference with burial services and visitor use of the site.

2.5 Habitats & species to be monitored

Primary monitored species

The primary species of concern is *Quercus robur* (pedunculate oak; Dutch: zomereik). All seven permanently monitored trees are individuals of this species. *Q. robur* is a long-lived, deciduous broadleaved tree of high ecological, cultural, and conservation value in the Netherlands. A study from 2019 states it as a keystone species supporting unique biodiversity; it is estimated that approximately 2,300 wildlife species such as, invertebrate species, lichens and fungi, and a wide range of cavity-nesting and insectivorous birds benefit from or rely on pedunculate oak and the health of these trees is important for the broader ecosystem [6]. The species prefers moist soil with a pH between 4.5 and 7 [5].

The vitality of the seven monitored oak trees is assessed through the full area of crown condition, foliar chemistry, and chlorophyll indicators described in Section 3.3.

No other tree species or vegetation types fall within the scope of this monitoring plan. The broader habitat, however, including the sandy soil within a heathland and woodland landscape, provides important context for interpreting the results. [7]

Habitat context

The monitored trees occur within a semi-natural woodland habitat managed as a nature burial ground. The canopy at Huis ter Heide is mixed, with *Quercus robur* co-occurring with Scots pine (*Pinus sylvestris*) and scattered silver birch (*Betula pendula*). The understorey is characterised by species typical of nutrient-poor, acidic sandy soils on the Utrechtse Heuvelrug, including bilberry (*Vaccinium myrtillus*), wavy hair-grass (*Deschampsia flexuosa*), and bracken (*Pteridium aquilinum*). Ground-layer vegetation is generally sparse beneath the denser canopy areas but becomes more diverse in lighter gaps.

The habitat type corresponds broadly to the European Union Habitats Directive Annex I habitat type 9190 (Old acidophilous oak woods with *Quercus robur* on sandy plains), which is characterised by acidic, nutrient-poor soils, oak as the dominant canopy species, and an understorey community adapted to low nutrient availability [8]. This habitat type is of conservation interest in the Netherlands and supports a range of specialist species.

The presence of declining or dead-wood-bearing oak trees at the site contributes to habitat value for saproxylic (dead-wood-dependent) invertebrates, cavity-nesting birds such as the great spotted woodpecker (*Dendrocopos major*), and fungal communities associated with old oak. Visual assessments carried out under this protocol record fungal fruiting bodies, bark lesions, and insect feeding damage where observed, providing contextual information on biotic stressors and ecological interactions that may be relevant to interpreting trends in oak vitality.

2.6 Timeframe

The monitoring is structured around the growing season of *Quercus robur* and is designed to be repeated annually. A single growing season provides an initial assessment of tree condition and treatment differences, but detecting the long-term effects of soil amendments such as bokashi on tree vitality requires a multi-year time series. A minimum of three to five growing seasons of standardised monitoring is recommended before robust conclusions on treatment effectiveness can be drawn.

2026 monitoring season

The 2026 growing season is the first complete, standardised monitoring season conducted under this protocol. Three monitoring rounds are scheduled at phenologically distinct stages of oak development, with a minimum of six weeks between rounds to allow ecologically meaningful change to be detected. An optional fourth round may be added at the discretion of the site manager.

- **Round 1 - Baseline** (early to mid-May 2026): conducted at full leaf expansion, providing the baseline dataset for the season.
- **Round 2** (late June to early July 2026): conducted during the peak growing season, when leaf physiological activity is at its highest.
- **Round 3** (late August to mid-September 2026): conducted in the pre-senescence phase, capturing conditions as the growing season draws to a close.
- **Round 4 - Optional** (late September to October 2026): a supplementary visual assessment at the onset of leaf ageing, if resources permit.

Multi-year continuation

From 2027 onwards, the same three-round monitoring schedule is to be repeated each growing season. Repeating the protocol using identical methods, timing, and tree identifiers ensures that measurements from successive years are directly comparable, enabling trend analysis at both the individual tree level and the stand level.

The multi-year time series will make it possible to determine whether any differences observed between the treated and reference trees in 2026 persist, strengthen, or diminish over time. It will also capture inter-annual variation driven by climatic factors such as drought or an exceptionally wet spring; this must be considered when interpreting treatment effects. When drone-based NDVI imagery is collected in multiple years, the spatial data layer will likewise accumulate into a time series that can show large-scale trends in canopy greenness across the burial ground.

It is recommended that the monitoring protocol and data are formally reviewed by the site manager at the end of each growing season, and that the protocol is updated if methods, tree composition, or management practices change. Any updates should be documented with the date and reason for the change.

3.0 Protocols/ methods to be applied

3.1 Frequency and responsibilities

Leaf sampling is conducted a minimum of three times per growing season. Monitoring rounds are timed to key phenological stages of *Quercus robur* and should be separated by at least six weeks to allow for ecologically meaningful change to be detected. An optional fourth round may be added at the decision of the site manager.

Recommended monitoring schedule:

Table 1 The recommended monitoring schedule for *Q. robur*

Round	Target period	Phenological context	Primary activities
Round 1 (Baseline)	Early–mid May	Full leaf expansion	Tree inventory check; PhotoFolia readings; leaf collection; visual assessment
Round 2	Late June – early July	Peak growing season	PhotoFolia readings; leaf collection; visual assessment
Round 3	Late August – mid-September	Pre-senescence	PhotoFolia readings; leaf collection; visual assessment
Round 4 (Optional)	Late September – October	Senescence onset	Supplementary visual assessment if resources permit

The monitoring plan is designed so that it can be carried out by two trained observers with no laboratory equipment required for field activities. Laboratory analysis of leaf nutrient content is conducted by a contracted analytical laboratory and does not require on-site expertise.

Recommended roles and responsibilities:

Table 2 The recommended roles and responsibilities for the monitoring area

Role	Recommended background	Key responsibilities
------	------------------------	----------------------

Field observer (lead)	Ecological field experience or equivalent training	Coordinates field sessions; completes tree inventory records; leads PhotoFolia measurements; manages sample labelling and chain of custody
Field observer (second)	Basic ecological or horticultural background	Assists with leaf collection; independently records visual crown assessment scores; photographs crown from second position
Site coordinator	Site management staff	Schedules access and site logistics; arranges transport of leaf samples to laboratory on the day of collection; maintains equipment inventory
Data manager	Any staff with basic spreadsheet proficiency	Enters field data into monitoring spreadsheet of each site visit; reviews entries against field sheets; flags anomalies for follow-up
Laboratory analyst	Contracted analytical laboratory	Receives, processes, and analyses dried leaf samples for nutrient content (MP-AES); returns results within agreed turnaround time

It is recommended that at least one field observer attends a practical orientation session with the monitoring protocol before the first independent round. All personnel handling samples should be familiar with the chain of protection requirements described in Section 3.2.

3.2 General oak leaf sampling protocol

Pre-sampling preparation

Before each field sampling session, all required materials should be checked and packed. Transport must be arranged so that leaf samples reach the laboratory on the same day as collection or are refrigerated within two hours if same-day transport is not possible.

Site and plot selection

The monitoring site is Natuurbegraafplaats Huis ter Heide, located in the Netherlands. Within this site, a fixed set of seven pedunculate oak (*Quercus robur*) individuals composes the permanent monitoring plot. The plot boundaries are fixed; no new trees should be added to or removed from the monitoring set without updating this protocol.

Each tree is georeferenced using a GPS-enabled device with a minimum positional accuracy of ± 3 m. Recorded coordinates and the accessible sampling position for each tree are documented in the Tree Inventory Sheet (Section 3.3) and must remain consistent between monitoring rounds to ensure the comparability of results.

Tree selection and tagging

The seven monitored trees are permanently identified by tree tags numbered T01 through T06. Tags must be inspected at the start of every field session and replaced immediately if lost or damaged, using the same numbering.

The tree ID, GPS coordinates, diameter at breast height (DBH), estimated crown height class, and soil treatment history are recorded in the Tree Inventory Sheet (Section 3.3) at the time of first tagging and updated whenever site conditions change (e.g., if a tree is chopped, or a new treatment is applied).

Of the seven trees, three have received annual bokashi soil applications since 2022 (treatment trees: FT01, FT02, FT03) and four serve as untreated reference trees (NFT04, NFT05, NFT06, HT07). This designation should remain fixed to allow long-term trend comparison, unless site management decisions require a change.

Leaf sampling procedure

Leaf samples are collected from the lower to mid-crown zone of each tree. The preferred method is direct hand-picking from the lowest accessible branches. Where necessary, a telescopic pole pruner (maximum reach approx. 4 m) may be used.

For each tree, approximately 260 leaves are collected, divided into four labelled sample bags of approximately 65 leaves each. Leaves are selected from four different branch positions distributed around the crown (N/S/E/W).

Only leaves meeting all of the following criteria should be collected:

- Fully expanded and fully green
- Sun-exposed (not deep shade leaves from inside the crown)
- Free from visible damage, insect damage, or fungal infection
- Not yellowing or senescing

Before taking leaves for laboratory analysis, PhotoFolia chlorophyll readings must be taken from four different branch positions distributed around the crown (north, west, east, south) in triploids (see Section 3.3). Readings are recorded immediately once the leaf is removed from the branch to avoid measurement artefacts.

Collected leaves are placed immediately into labelled resealable bags, kept cool and shaded, and transported to the laboratory on the same day. If same-day transport is not possible, samples must be refrigerated within two hours of collection and delivered no later than 24 hours after sampling.

Field documentation and chain of custody

A completed field data sheet (Tree Inventory Sheet and PhotoFolia Readings and Leaf Sampling Sheet; see Section 3.3) is required for every tree at every monitoring round. Completed paper sheets should be photographed immediately after each session as a digital backup, and the originals retained in a dedicated monitoring folder.

All sample bags must be labelled before leaving the field with:

- Tree ID (e.g., T03)
- Monitoring round and year (e.g., R2-2026)
- Sample bag sequence (A, B, or C)
- Date of collection (DD-MM-YYYY)
- Collector initials

A chain of custody record must be completed for each batch of bags delivered to the laboratory, noting: tree ID, round, sample type (leaf), date and time of collection, collector name, date and time of laboratory receipt, and receiving analyst name.

All field data should be entered into the site monitoring spreadsheet within 48 hours of each visit. A second person should check the entries against the original field sheets within one week to confirm accuracy before results are used to inform any management decisions.

3.3 Field data recording templates

Tree Inventory Sheet (example format)

The Tree Inventory Sheet is completed once at the time of initial tagging and updated at the start of each new monitoring season, or whenever relevant attributes change. Height class: 1 = below 10 m, 2 = 10–15 m, 3 = 15–20 m, 4 = 20–25 m, 5 = above 25 m.

Table 3 Tree Inventory Sheet (example format)

Tree ID	GPS Lat (WGS84)	GPS Lon (WGS84)	DBH (cm)	Height class	Bokashi treatment	Date last updated	Updated by	Notes
T01					Yes (since 2022)			
T02					Yes (since 2022)			
T03					Yes (since 2022)			
T04					No (reference)			
T05					No (reference)			

T06					No (reference)			
T07					No (reference)			

PhotoFolia Readings and Leaf Sampling Sheet

About PhotoFolia

PhotoFolia is a free smartphone application for non-destructive, field-based estimation of leaf chlorophyll content. The app is available for iOS and Android. It uses photographs of individual leaves taken against a printed colour-standard card (supplied with the application) to calculate a Leaf Greenness Value (LGV). The colour card corrects for variable lighting conditions in the field, producing a consistent, quantitative index that correlates with leaf chlorophyll concentration [9].

PhotoFolia readings serve as a practical field indicator of tree leaf health and can be taken by any observer with a smartphone, without specialist equipment. For best comparability between rounds, always use the same smartphone model and the same colour card version. Replace the colour card if it becomes faded or damaged.

How to use PhotoFolia in the field:

1. Open the PhotoFolia app and confirm successful calibration of the colour card at the start of each session. If calibration fails, move to a shaded or uniformly lit area and try again.
2. Place the colour-standard card flat on a stable surface. Ensure the card and the leaf are in the same plane and both fully lit; avoid direct harsh sunlight or deep shadow.
3. Hold the target leaf gently against the card without folding it. Do not cover the printed colour patches with the leaf.
4. Frame the leaf in the app camera view and capture the image. The app will detect the card and display the LGV automatically.
5. Record the LGV immediately on the data sheet.
6. Repeat for 12 leaves per tree, chosen from different positions (N/S/E/W) around the accessible crown.
7. At the end of each field session, export the session data from the app as a CSV file and save it to the site monitoring folder.

For more detailed instructions, refer to the website: [PhotoFolia Colourworker User Guide](#) [10]

Session information :

Table 4 Session information (example format)

Field	Details
Date	
Monitoring round and year	
Tree ID	
Observer name	
Smartphone model used	
Colour card ID / version	
Weather / lighting conditions	

Leaf greenness readings:

Table 5 Leaf greenness readings (example format)

Leaf no.	Branch position (N/S/E/W)	PhotoFolia LGV	Notes
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			
11			
12			
Mean LGV			

Leaf sample collection record:

Table 6 Leaf sample collection record (example format)

Crown transparency	Crown transparency	Crown transparency	Crown transparency
[TreeID]-R[n]-A	~65		
[TreeID]-R[n]-B	~65		
[TreeID]-R[n]-C	~65		
Total	~195		

3.4 Visual assessment protocol

The purpose of this protocol is to allow for consistent monitoring of the health of the seven Oak trees by any two observers on site. For this protocol, no specialist equipment except binoculars and this form will be needed. The protocol will consist of 4 different assessments, supporting information and an assessment table to report the findings per test, as well as additional damage remarks.

These assessments are preferably conducted during the full leaf out of the *Quercus Robur*, usually late May until late July, on a day when the sky is clear.

Materials

Minimum equipment list (per assessment session):

- Binoculars (recommended magnification: 8x or 10x)
- This protocol document (printed out is recommended)
- Pencil/ pen
- Camera or smartphone with a good functioning camera
- Optional: measuring tape to mark the observer positions to the tree

Observer positioning

8. The observer should be positioned at approximately one tree-height away from the trunk (10-20 m). Mark this position to ensure consistency during subsequent assessments.
9. Two observers stand at approximately 90 degrees apart, around the tree base, to make independent observations before agreeing on the final score.
10. Use binoculars for the crown observations and photograph the crown from each observer's perspective with a consistent zoom.
11. Perform the 4 assessments presented in this protocol.

Crown defoliation

Crown defoliation is the estimated percentage of reduced leaf density relative to a healthy reference tree. Follow the IPC forest method by following the table and diagram below to assign a defoliation class.

Table 7 Overview of the different defoliation classes [11]

Class	Defoliation	Description
0	0–10%	No or very slight defoliation; healthy crown
1	>10–25%	Slight defoliation; minor stress indicators
2	>25–60%	Moderate defoliation; crown noticeably thinned
3	>60–99%	Severe defoliation; crown heavily damaged
4	100%	Dead crown (code 99 if tree still alive)

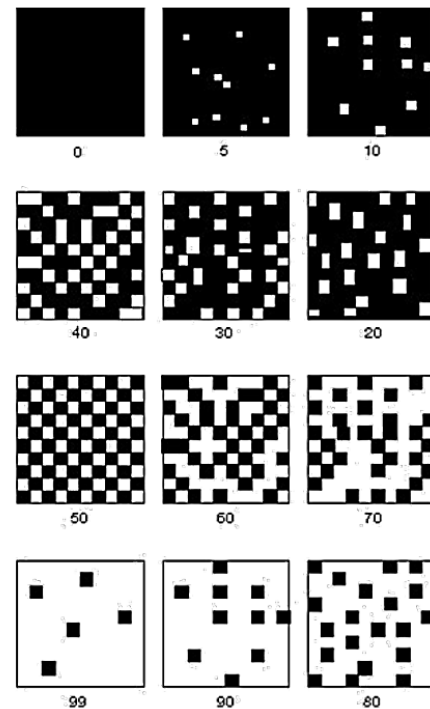


Figure 1 Overview of the different levels of defoliation [11]

The table summarises each defoliation class and its defoliation percentage range, as well as a brief description to ensure consistent scoring between assessments. The figure shows increasing levels of defoliation with black and white silhouettes of what the oak leaf dispersity may look like. The black dots represent the amount of leaf density on the tree.

Assessment

Use the reference images and defoliation table, which class compares to the overall density of the crown. Working from class 0 (full healthy crown) downward until the most representative match is found. When in doubt between two classes, choose the class with the higher severity of defoliation. Report this in the table at the end of this protocol.

Crown transparency

Crown transparency is the estimated proportion of skylight visible through the foliated crown. The crown transparency is measured in steps of 5%. The assessment is performed by mentally drawing out a 2D silhouette of the tree and estimating the visible sky within the foliated zone, with bare branches being treated as if transparent [12].

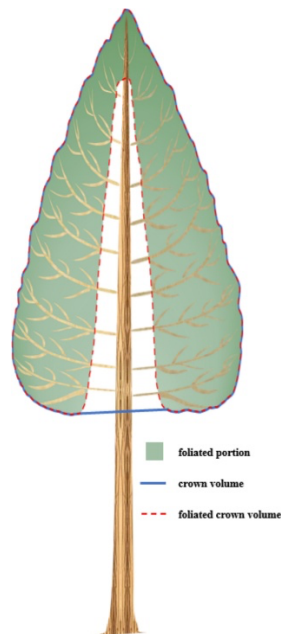


Figure 2 Schematic overview of the definitions Foliated portion, Crown volume and Foliated cover volume [12]

The Figure illustrates the vertical cross-section of a tree, providing a visual of the meaning of the concept of the foliated portion of a tree.

Assessment

Imagine a 2D silhouette of the tree by mentally drawing the outer boundary of the foliated zone. Exclude the area below the lowest branches and treat bare branches as invisible or transparent. Now, estimate (in %) the amount of sky that is still visible within the silhouette. Round this estimated percentage to the nearest 5% (for example, an estimated 23% is reported as 25%). Report this in the table at the end of this protocol.

Crown dieback

Dieback refers to recent dead branches and fine branch tips in the upper and outer crown zones, recorded in 5% increments, using the same table as for crown defoliation. Lower crown dead branches are excluded.

Assessment

Estimate how much recent dieback, fine dead branch tips and dead branches the tree has, ignoring the dead branches in the lower crown. Use the table from the defoliation assessment and estimate the percentage of crown dieback. Round this estimated percentage to the nearest 5% (for example, an estimated 23% is reported as 25%). Report this in the table at the end of this protocol.

Epicormic growth

Epicormic growth is a stress-response in trees, resulting in latent buds on the trunk and main branches. Epicormic growth is measured using a 3-point scale.

- 0 – No epicormic shoots visible
- 1- Scattered shoots present, not dominating
- 2 – abundant shoots, clearly dominating the trunk and lower leaf crown.

Assessment

Look at the whole circumference of the tree trunk for clusters of small, thin shoots that are growing directly out of the trunk. Also, look at shoots from the main branches that do not form normal branch junctions. Give the amount of epicormic growth a score as presented above[13].

Additional damage observations

Any of the following remarks noticed should be recorded: presence of fungal activity, bark cankers/lesions, mechanical damage, insect damage, leaf discolouration, fruit production level (absent/ scarce/ common/ abundant) [13].

Data recording and visual assessment data sheet

For this visual assessment to be successful, the visual data sheet should be completed for each tree in each assessment. Both observers record independently, deciding together on the final score per tree. All photographs should be labelled with the tree ID, position and observer.

Table 8 Table to report the observed assessment data (example format)

Field	Observer 1	Observer 2
Date		
Tree ID		
Observer name		
Observer position (°)		
Weather conditions		
Crown defoliation class (0–4)		
Crown defoliation % estimate		
Crown transparency %		
Crown dieback %		
Epicormic growth (0–2)		
Epicormic shoot location		
Damage agent(s) identified		
Photo reference numbers		
Agreed final defoliation class		
Remarks		

3.5 Leaf Monitoring Protocol

In this experiment, we will use a non-destructive approach to investigate the leaf development process for seven individual *Quercus robur* plants during the vegetation period. To do this, we need to take photos of the leaves in spring and observe their changes throughout the summer and autumn seasons. In other words, this approach allows us to assess the development of leaves without physically approaching the plants.

To conduct an experiment, you need to select your individual specimens of *Quercus robur* and allocate each plant its identifier number. You also need to choose one point on the ground where you will observe each tree throughout the growing season. Then you need to take a photo of the selected part of the canopy.

Monitoring Procedure

Next, photographs must be taken from time to time (weekly or every two weeks). It is necessary to take pictures from one point with the help of a camera and under similar conditions. Images must include a clear view of a cluster of leaves to evaluate the state of the foliage.

Data Recording and Analysis

The resulting photos must be carefully annotated with tree identifiers and dates of observation. Comparing the resulting images, one can assess the growth rate of the leaves and their condition. The stages of development of oak leaves correspond to the stages of vegetation of *Quercus robur*.

Table 9 Data recording of *Q. robur* (example format)

Date	Tree ID	Leaf Development Bud/Emerging/ Full	Leaf Growth Very Small/Small/Medium/Large	Leaf Color Light green / Green / Dark green / Yellow / Brown	Visible Damage/Stress None / Mild / Moderate / Severe
	1				
	2				
	3				
	4				
	5				
	6				
	7				

3.6 Laboratory standard operating procedures

3.6.1 Sample preparation [5], [14]

Representative, uncontaminated, and homogeneous *Quercus robur* leaf samples are required to accurately capture within-tree variability. This is achieved through a structured

sampling design, consistent leaf selection, and careful handling and preparation prior to analysis.

Materials and Equipment

- Pre-labelled sample containers
- Permanent markers
- Drying oven (adjustable 40–105 °C)
- Desiccator with silica gel (optional)
- Mechanical grinder or blender
- Sieves
- Tweezers
- Spatulas

Sample logging & storage conditions

Each sample is assigned a unique identification code to ensure traceability throughout the analysis process. For every collected sample, relevant metadata are recorded, including the sampling location, date and time of collection, sample type and condition, and the name of the collector. This systematic documentation ensures consistency, enables accurate data interpretation, and allows any variations in results to be linked back to specific sampling conditions.

Fresh leaf samples are stored at 4 °C and analysed within a maximum of 24 hours to minimise biochemical changes and degradation. Throughout storage, care is taken to avoid cross-contamination and exposure to dust or other external contaminants, ensuring the integrity and reliability of the samples.

Drying procedure

- Spread leaves individually to avoid overlapping (oak leaves retain moisture).
- Add weights on the leaves to avoid movement.
- Dry at 65 °C for 48 hours.
- Ensure crisp texture before grinding (leaves should snap cleanly when broken).
- Verify constant weight: the mass change over the last 24 hours should be less than 1 %.

Grinding and homogenisation

- Grind each tree's dried leaves to a fine powder.
- Mix thoroughly to ensure homogeneity within each tree's powder.

- Store subsamples in airtight containers within a desiccator.
- Re-homogenise (roll and stir for ~30 seconds) before each weighing step, as powder can settle and stratify in storage.

3.6.2 Acid digestion for MP-AES analysis [14], [15], [16]

Accurate determination of elemental concentrations in oak leaf samples requires complete digestion of the plant matrix followed by instrumental analysis. In this method, dried and homogenised leaf samples are subjected to acid digestion using concentrated nitric acid in a microwave digestion system to break down organic material and release elements into solutions.

Materials and Equipment

- Nitric acid (HNO₃, 65 %, trace-metal grade)
- Deionised water (MilliQ)
- Microwave digestion system
- Digestion vessels with caps
- Volumetric flasks (50 mL, Class A)
- Analytical balance (4-decimal)
- Fume hood

Microwave digestion procedure

- Weigh 0.25 g (± 0.005 g) of dried, ground leaf sample directly into a labelled digestion vessel. Record the exact mass to four decimal places.
- In the fume hood, add 10 mL of concentrated HNO₃ slowly down the side of the vessel.
- Seal the vessels with their caps according to the microwave's specification.
- Load vessels into the rotor in a balanced configuration.
- Run the digestion programme according to the microwave manufacturer's recommended protocol for plant tissue.
- Allow the vessels to cool to below 60 °C inside the microwave before opening the door.

3.6.3 Preparation of Samples [17], [18], [19]

This protocol describes the steps taken once the digestion vessels have come out of the microwave. It covers cooling, visual inspection, quantitative transfer to a volumetric flask, filtration, and the 1:2 dilution required before MP-AES analysis. The protocol applies to all sample types processed in the same digestion batch: leaf samples, method blanks, and any quality control samples.

Materials and Equipment

- Deionised water (MilliQ, $\geq 18.2 \text{ M}\Omega\cdot\text{cm}$)
- 50 mL Class A volumetric flasks with stoppers (one per digested vessel)
- 15 mL polypropylene autosampler tubes with screw caps (two per digested vessel — one for the filtered digest, one for the diluted aliquot)
- Disposable 5 or 10 mL plastic syringes (one per sample)
- 0.45 μm PTFE syringe filters (one per sample)
- Funnels (one per sample)
- Disposable polypropylene pipette tips (fresh tip for each transfer)
- Pasteur pipettes for meniscus adjustment
- Micropipettes covering the 1–10 mL range
- Permanent marker (alcohol-resistant) and printed labels

Cooling and opening of vessels

- Transfer the vials to a fume hood. Vessels must not be opened outside the fume hood.
- Unscrew the cap of one vessel at a time, slowly, with the vessel pointed away from your face. If a noticeable release of pressurised vapour occurs, wait a few seconds before continuing.

Quantitative transfer and dilution to 50 mL

- Label clean 50 mL volumetric flasks with the same identifier as the digestion vessel
- Carefully pour the contents of the digestion vessel into the corresponding labelled volumetric flask, using a funnel to avoid spillage.
- Rinse the inside of the empty digestion vessel with approximately 2 mL of MilliQ water, swirl to wash the walls, and pour the rinse into the volumetric flask. Repeat this rinse two more times (three rinses total).

- Rinse the digestion vessel cap and inside threads with MilliQ water into the flask if any residue is visible on them.
- Top up the volumetric flask with MilliQ water until the meniscus approaches the 50 mL graduation mark.
- Use a clean Pasteur pipette to add the final drops, aligning the bottom of the meniscus exactly with the 50 mL mark at eye level.
- Stopper the flask and invert 10 times to mix thoroughly. Do not shake.

The undiluted digest at this stage contains approximately 7.8 % HNO_3 (v/v) from the 10 mL of concentrated acid used during digestion.

- Label clean 10 mL plastic tube with the sample identifier as the volumetric flasks.
- Draw approximately 10 mL of the digest from the volumetric flask into the tubes.
- Cap the tubes tightly.
- Retain the remaining volume in the volumetric flask, stoppered, as a secondary backup until the analysis is verified complete.

1:2 dilution for MP-AES analysis

The MP-AES instrument tolerates a maximum acid concentration of 5 % (v/v). The undiluted digest at 7.8 % HNO_3 exceeds this limit and must be diluted by a factor of 2 with MilliQ water before measurement. The resulting solution is at approximately 3.9 % HNO_3 , matching the matrix of the calibration standards.

- Label the same amount of plastic tubes with the sample identifier and the suffix "-D" (for diluted)
- Using a clean micropipette tip, pipette 5.0 mL of the filtered digest into the labelled -D tube.
- Using a fresh tip, pipette 5.0 mL of MilliQ water into the same -D tube. Do not use acid as the diluent.
- Cap the tube and invert 10 times to mix.

The diluted sample (~3.9 % HNO_3) is now ready for MP-AES analysis.

Storage

- Diluted samples for immediate analysis: transfer to the MP-AES autosampler rack and proceed with the measurement sequence within 2 hours of dilution.
- Diluted samples for delayed analysis: store at 4 °C in a refrigerator, tightly capped, for up to 1 week.

Calculation parameters

The following parameters are used in the final calculation of leaf concentrations from MP-AES readings (mg/L).

Table 10: Parameters for calculations

Parameter	Value	Source
Final volumetric flask volume	0.050 L	Quantitative transfer step
Post-digestion dilution factor	2	1:2 dilution step
Sample dry mass (m)	Variable; in kg	Recorded during weighing for digestion (Section 3.6.2)

The leaf concentration (mg/kg dry matter) is calculated per replicate as:

$$C_{\text{leaf}} (\text{mg/kg DM}) = (C_{\text{measured}} - C_{\text{blank}}) \times 0.050 \times 2 \div m_{\text{sample}} (\text{kg})$$

Where C_{measured} is the MP-AES reading in mg/L, C_{blank} is the average of the method blank readings, and m_{sample} is the exact dry mass of leaf powder weighed for that replicate.

3.6.4 Preparation of Calibration Standards [20], [21], [22]

This protocol describes the preparation of a seven-point multi-element calibration series (S0, S0.5, S1, S2, S3, S4, S5) for the simultaneous determination of eight elements (K, Ca, Mg, Mn, Al, Fe, Zn, Cu) in acid-digested oak leaf samples. The standards are matrix-matched to the diluted samples at approximately 3.9 % HNO_3 (v/v).

The procedure consists of three stages: first, preparation of a 100 mg/L multi-element working stock by combining 1 g/L single-element standards; second, dilution of the working stock into the main calibration series (S0–S5); and third, preparation of an additional low-concentration standard (S0.5) by dilution of S1.

Materials and Equipment

- 1 g/L single-element standards in 1 % HNO_3 matrix, one for each of: K, Ca, Mg, Mn, Al, Fe, Zn, Cu
- Concentrated HNO_3 (≥ 65 %, trace-metal grade)
- Deionised water (MilliQ, $\geq 18.2 \text{ M}\Omega \cdot \text{cm}$)
- 1 × 100 mL Class A volumetric flask (for the multi-element working stock)
- 7 × 100 mL Class A volumetric flasks (for S0, S0.5, S1, S2, S3, S4, S5)

- 8 × 15 mL polypropylene autosampler tubes with screw caps
- Micropipettes covering the 100–1000 µL and 1–10 mL ranges
- 10 mL graduated pipette or acid dispenser for HNO₃
- 25 mL and 50 mL graduated pipettes
- Disposable polypropylene pipette tips (fresh tip for each transfer)
- Pasteur pipettes for meniscus adjustment
- Dark plastic bottle (or amber glass) for working stock storage
- Permanent marker (alcohol-resistant) and printed labels

Preparation of multi-element working stock (100 mL at 100 mg/L of each element)

Final acid matrix: approximately 0.8 % HNO₃ (contributed by the 1 % HNO₃ matrix of the single-element standards; no additional acid is added at this stage).

- Label a clean 100 mL Class A volumetric flask: Multi-element stock 100 mg/L — [date] — [initials].
- Add approximately 20 mL of MilliQ water to the flask.
- Using a clean micropipette and a fresh tip for each transfer, pipette 10.0 mL (10,000 µL) of each of the eight 1 g/L single-element standards into the flask, swirling gently between additions.

Table 11 Preparation of multi-element working stock.

Order	Single-element standard	Volume
1	1 g/L K standard	10.0 mL
2	1 g/L Ca standard	10.0 mL
3	1 g/L Mg standard	10.0 mL
4	1 g/L Mn standard	10.0 mL
5	1 g/L Al standard	10.0 mL
6	1 g/L Fe standard	10.0 mL
7	1 g/L Zn standard	10.0 mL
8	1 g/L Cu standard	10.0 mL

- Inspect the solution visually. It must remain clear; a faint pale yellow tint from iron is acceptable. If cloudiness or precipitate forms, do not proceed — discard and remake the stock.
- Top up to the 100 mL graduation mark with MilliQ water, using a Pasteur pipette for the final drops to align the meniscus precisely at eye level.
- Stopper the flask and invert 10 times to mix thoroughly.

- Transfer the stock to a clean, labelled dark plastic bottle (or wrap the volumetric flask in foil). The label must include: contents, matrix (~0.8 % HNO₃), date of preparation, and operator initials.
- Store at 4 °C until use. The working stock is stable for up to 3 months when refrigerated and tightly sealed.

Preparation of calibration standards S0–S5

Final acid matrix in each standard: approximately 3.9 % HNO₃, matched to the diluted sample matrix prepared in Section 3.6.3.

- Label six clean 100 mL Class A volumetric flasks as S0, S1, S2, S3, S4, S5.
- Label six clean 15 mL polypropylene autosampler tubes with the matching labels.
- To each of the six flasks, add approximately 50 mL of MilliQ water.
- Using a dispenser or 10 mL graduated pipette, add 3.9 mL of concentrated HNO₃ to each flask.
- Using a clean micropipette tip for each transfer, add the appropriate volume of multi-element working stock to each flask according to the table below.

Table 12 Preparation of calibration standards S0–S5.

Standard	Target concentration (mg/L)	Volume of working stock
S0	0	0 µL (none added)
S1	1	1,000 µL (1.0 mL)
S2	5	5,000 µL (5.0 mL)
S3	10	10,000 µL (10.0 mL)
S4	25	25,000 µL (25.0 mL)
S5	50	50,000 µL (50.0 mL)

- Swirl each flask gently after the stock has been added.
- Top up each flask to the 100 mL graduation mark with MilliQ water, using a Pasteur pipette for the final drops.
- Stopper each flask and invert 10 times to mix thoroughly.
- Transfer approximately 10 mL of each calibration standard into its corresponding labelled polypropylene autosampler tube. Cap tightly.
- Keep the volumetric flasks stoppered as backup. The standards in the flasks may be stored at 4 °C and reused within 24 hours; for best results, prepare fresh standards on the day of MP-AES analysis.

Storage and stability

- The multi-element working stock is stable for up to 3 months at 4 °C in a sealed dark container.
- Calibration standards S0–S5 and S0.5 should be prepared fresh on the day of MP-AES analysis. If reuse is necessary, store at 4 °C and use within 24 hours.
- Dilute single-element solutions at 1 mg/L and below (S0.5 and S1) are particularly prone to wall adsorption losses over time, especially for Fe, Cu, and Zn; same-day preparation is strongly recommended.

3.6.5 MP-AES analysis [23], [24], [25]

Calibration Standards

Prepare multi-element calibration standards in an acid matrix matching the digested samples for each of the seven leaf batches individually. For every batch, a calibration series is prepared consisting of 0 (blank), 1, 5, 10, and 20 mg/L standards, with the concentration range adjusted if necessary to match the expected sample concentrations. Use 5 volumetric flasks (100 ml), pipettes and tips.

Analytical Procedure

- Start the MP-AES (or let the lab assistant do it) and allow the plasma to warm up for at least 30 minutes.
- Follow the start-up procedure provided by the MP-AES.
- Label and load calibration standards, blanks, and samples into the designated autosampler tube racks.
- In the software, select the elements to be measured. For each element, choose two wavelengths that do not interfere with any other measured wavelengths (the software flags interferences in red).
- Set the per-element calibration range so that each calibration curve is fitted only to the standards within the relevant concentration range (see table below).
- Insert all tubes into the designated tube racks of the MP-AES.
- Check that the MP-AES has recognised all tubes in the racks.
- Run the calibration sequence first. Verify that $R^2 \geq 0.999$ for every element before continuing to the samples.
- Run the blank + samples.

- When the MP-AES is done, transfer all data to a USB stick.

Quality Control & Data Processing

Quality control is ensured by evaluating recovery rates (acceptable range 90–110%), maintaining a relative standard deviation (RSD) below 5% for replicate measurements, and confirming that blank signals are negligible. Calibration curve linearity is verified with an R^2 value of at least 0.995, and mid-level standards are reanalyzed periodically to check instrument stability. For data processing, measured concentrations are corrected for any dilution factors applied during sample preparation and subsequently converted from milligrams per liter (mg/L) to milligrams per kilogram (mg/kg) based on the original sample mass. Final results are reported together with appropriate uncertainty estimates to reflect analytical precision and reliability.

3.6.6 Chlorophyll Extraction and Spectrophotometry [26], [27], [28]

Efficient extraction and quantification of chlorophyll pigments (chlorophyll a and b) provide insight into photosynthetic capacity and plant physiological status. Careful handling minimizes pigment degradation and reduces interference from phenolic compounds present in oak leaves. Oak leaves contain pigments alongside tannins and phenolics, which may interfere with extraction if not handled properly.

Equipment

- Centrifuge (3000–5000 rpm)
- Spectrophotometer (visible range)
- Glass cuvettes (1 cm path length)
- 80% acetone + OR 95% ethanol OR DMSO

Procedure (Optimized for Oak Leaves)

1. Take a 1 cm² fresh leaf tissue (avoid major veins if possible)
2. Put these in 10 cm plastic tubes
3. Label all tubes
4. Add 2.4 ml of acetone, and 0.6 ml of demiwater.
5. Keep samples in darkness (wrap tubes in foil)
6. Store the tubes in a tube rack in the fridge for 24 hours.
7. The liquid should turn green. If not, use a sharp object to smash the leaves in the tube. Make sure this sharp object is cleaned with acetone.
8. Wait another 30 minutes.
9. Centrifuge for 5 minutes at 3000 rpm.
10. In the meantime, make sure the glass cuvettes are clean. (Clean with acetone).
11. Pipette the sample in the glass cuvette. Do not touch the clear sides of the cuvette with your hands.
12. Measure absorbance at:

- 663 nm (Chl a)
- 645 nm (Chl b)

Calculations

- $\text{Chl a} = 12.7(A_{663}) - 2.69(A_{645})$
- $\text{Chl b} = 22.9(A_{645}) - 4.68(A_{663})$
- $\text{Total Chl} = \text{Chl a} + \text{Chl b}$

3.6.7 Hach Lange tests protocols [22], [29]

Leaf nutrient analysis provides insight into plant nutritional status and nutrient uptake efficiency. In this method, acid-digested leaf samples are analyzed for nitrogen (as nitrate and ammonium) and phosphorus (as phosphate) using colorimetric cuvette tests from Hach Lange. These tests quantify dissolved nutrient ions based on light absorption after specific chemical reactions, measured using a spectrophotometer. The resulting concentrations are used to assess plant nutrient availability and physiological status.

Equipment

- Acid-digested leaf extracts (prepared in advance)
- Hach Lange nitrate cuvette test kits (P)
- Hach Lange nitrate cuvette test kits (N)
- Hach Lange spectrophotometer (DR6000 series)
- Micropipettes and tips
- Cuvette cleaning cloth
- Deionized water (blanks and dilution)
- Timer

Sample & Blank Preparation

All acid-digested leaf samples are first allowed to cool to room temperature and are thoroughly mixed to ensure homogeneity before analysis. If necessary, samples are diluted with deionized water to bring nutrient concentrations within the measurable range of the test kits. Each sample is then prepared for analysis by ensuring it is free of particulates and properly homogenized prior to pipetting into the cuvette tests.

A blank sample is prepared by filling a clean cuvette with deionized water, which serves as the reference for zero absorbance. The cuvette is carefully wiped with a lint-free cloth to remove fingerprints or residues before being placed into the spectrophotometer. This blank measurement is used to calibrate the instrument and correct for background absorbance prior to analyzing the actual samples.

Total Nitrogen (N) determination

1. Add in quick succession to a dry reaction tube: 0.5 mL sample, 2.0 mL solution A, 1 tablet B.
2. Close immediately the reaction tube. Do not invert.
3. Heat immediately. For the (if available) HT 200 S: in the standard program HT for 15 minutes. Thermostat: for 30 minutes at 120 °C (248 °F).
4. Allow to cool to room temperature.
5. Invert a few times.
6. Slowly pipet into the Cuvette Test: 0.5 mL of digested sample
7. Slowly pipet 0.2 mL of solution D.
8. Immediately close the cuvette and invert a few times until no more streaks can be seen
9. After 15 minutes, thoroughly clean the outside of the cuvette and evaluate.
10. Insert the cuvette into the cell holder.

Phosphate (P) determination

1. Carefully remove the protective film
2. Unscrew DosiCap Zip from cuvette.
3. Pipette 0.5 mL sample carefully.
4. Immediately screw the
5. DosiCap Zip back on ribbed side up.
6. Shake the cuvette vigorously.
7. Heat immediately. Thermostat: for 60 minutes at 100° C (212° F) or for 30 minutes at 120° C (248° F).
8. Let cool to room temperature. IMPORTANT: Check if the cap is still securely sealed after cooling
9. Shake the cuvette vigorously
10. Unscrew DosiCap Zip from cuvette.
11. Pipette 0.2 mL of reagent B into the cooled cuvette. Immediately close the reagent B bottle after use.
12. Screw a grey Dosicap C onto the cuvette.
13. Invert several times until the lyophilisate is completely dissolved.
14. After 10 minutes, invert the cuvette again, clean the outside thoroughly and measure
15. Insert the cuvette into the cell holder.

Calculations and quality control

Measured concentrations obtained from the spectrophotometer are first corrected for any dilution factors applied during sample preparation. These values are then converted from mg/L to mg/kg dry weight using the sample's extraction volume and original dry mass. Quality control is ensured by measuring blank samples regularly to check for contamination

or instrument drift, and by repeating at least 10% of all samples to assess analytical precision. In addition, cuvettes are inspected for cleanliness and scratches before use, and the spectrophotometer is checked for proper calibration prior to each measurement session.

3.7 Quality assurance and quality control documentation

This protocol defines the quality assurance and quality assessment framework. This protocol covers field sampling, laboratory analysis, and data management. Quality assurance refers to the planned system set out to ensure quality, whereas quality control checks the application by verifying the application of the methods.

3.7.1 Field sampling

Field sampling requires a chain of custody for every collected batch; it includes a record of the tree ID, sample type, collector name, and collection date and time. Approximately 200 leaves will be collected per tree (65 leaves per sample, 3 samples). These leaves are immediately labelled in bags and transported to the laboratory on the same day or refrigerated and transported to the laboratory within 24 hours of sample collection [5].

For a field sampling, the following quality assurance procedure steps should be followed:

1. Before leaving for the site for sample collection, confirm that all the required equipment is present and functional. This includes but is not limited to: binoculars, camera/ smartphone, printed out assessment tables, labelled sample bags, pen/ pencil, and a cool box if the analyses are expected to take long.
2. On arrival at the site, record in the logbook: the date, start time, present team members and weather conditions of the day. Take note of the cloud cover, signs of recent rainfall and wind (direction).
3. Before sampling, identify and verify the tree ID in Google Maps or ArcGIS
4. During field sampling, only collect the fully expanded leaves from the mid crown, with no sign of disease or damage.
5. After the collection of the approximately 200 leaves, the leaves must be immediately placed in a labelled brown paper bag. The paper bags must be stored upright and in a cool box during long sample days.
6. Complete the chain of custody as mentioned above for each sampling batch.
7. Note any inconsistencies from the sampling protocol. This includes but is not limited to: equipment malfunctions, inaccessible branches, and unexpected weather

conditions during or before the sample day. In case of a deviation from the expected events, the team should decide as a group whether the samples can still be used or discarded.

3.7.2 Laboratory analysis

Sample drying

The fresh leaves are placed in single (slightly overlapping) layers in the laboratory's ventilated oven and dried at 65 degrees Celsius for 24 to 48 hours. The leaves are considered dry when brittle and snap cleanly. The constant mass is confirmed by checking that the mass change over 24 hours is less than 1% [30].

For sample drying, the following quality assurance steps should be followed:

- Chain of custody: In the laboratory, first check the paper bags with leaves for the chain of custody requirements and note this down on the data logging table/ sheet before starting.
- Cleaning: If the leaves show visible signs of dirt, this should be gently rinsed with deionised water and blotted dry with a lint-free paper towel.
- Label drying trays: lay the leaves on properly labelled drying trays containing tree ID, laboratory ID and the date.
- Check the oven settings: Before placing the trays in a ventilated drying oven at 65 degrees Celsius, check the oven's settings and record the date and time in the laboratory journal.
- Check on samples: check on the samples after 24 hours. If the leaves snap clean when broken in half, they are dry.
- Desiccator: to ensure proper drying without moisture reabsorption, allow the leaves to cool down in the desiccator for 30 minutes.

Chlorophyll analysis

PhotoFolia chlorophyll measurement uses smartphone camera images of individual leaves and compares these with the provided colour standard cards.

For the chlorophyll measurement analysis, the following quality assurance steps should be followed:

- Colour standard calibration: confirm successful calibration when starting each session. If the calibration fails, repeat the process in different lighting.

- Duplicate leaf measurements: For 3 leaves per tree, two independent photographs should be taken to compare values. These values should not vary by more than 10 units. If the values vary by more than 10 units, check for consistent lighting.
- Cross-session consistency: the per-tree mean PhotoFolia value from round 1 serves as a baseline for the rest of the rounds. Changes of more than 20 units should be flagged for further investigation.
- Cross-validation with nutrient: after the chlorophyll analysis with PhotoFolia, the values are plotted against leaf nitrogen concentrations to assess if these values correlate as expected. R above or equal to 0.60 is the target value for the field proxy validation.

MP-AES analysis

The MP-AES analysis requires a calibration for every sample batch to ensure quality. All calibration curves must yield an R^2 value of at least 0.998 before sampling analysis proceeds.

The table below must be completed per batch before the analysis continues. Triplicate digestions are performed for every tree.

Table 13 MP-AES calibration table (example format) [13]

Element	Wavelength (nm)	Calibration range (mg/L)	Calibration equation	R^2	MDL (mg/kg DM)	Accepted? (Yes/No)
P	213.617	0–20.0			10	
K	766.490	0–20.0			15	
Ca	317.933	0–20			5	
Mg	285.213	0–20			5	
Fe	259.940	0–20			2	
Mn	257.610	0–20			1	
Zn	213.856	0–20			1	
Cu	324.754	0–20			0.5	

Table 13 outlines the steps to be completed for the MP-AES calibration. The wavelengths and calibration ranges are fixed settings; the empty columns are to be completed per sample batch.

Hach Lange cuvette tests (LCK 238 & LCK 348)

The Hach Lange colourimetric cuvette tests quantify total nitrogen (LCK 238, range 0.5–15.0 mg/L N) and reactive phosphate (LCK 348, range 0.15–5.00 mg/L P) in acid-digested leaf extracts using a DR6000 series spectrophotometer. Results are used in conjunction with

ICP-OES data to confirm foliar N and P status. The quality assurance steps below apply to every analysis session [22], [29].

Blank measurement

A blank cuvette with deionised water needs to be run at the start of every analysis session. After every 10 samples, using the same procedure, a blank should be read below the kit's detection limit.

For the measurement, the following quality assurance steps should be followed:

- Fill a cuvette with deionised water and wipe the outside with a lint-free cloth to remove fingerprints or residues.
- Process the blank through the test procedure for both LCK 238 and LCK 348.
- If the blank reads above the detection limit, we discard the cuvette, prepare a blank with a new batch of deionised water and repeat. If the second blank also fails, we suspend the analysis session. Identify the source of contamination before continuing.
- Note any failures, including the action taken.

Sample preparation and range check

For sample preparation and the range check, the following quality assurance steps should be followed:

- Allow all acid-digested leaf extracts to cool to room temperature and mix thoroughly before pipetting.
- If a sample is expected to exceed the kit measuring range based on results or high-concentration digests, we dilute it with deionised water before analysis and record the dilution factor.
- Inspect each cuvette before use. Cuvettes with scratches, deposits, or cloudiness are discarded; these will affect absorbance readings.
- Confirm that the reagent lot numbers are recorded and that the expiry dates of both LCK 238 and LCK 348 kits have not passed.

Duplicate measurements

We need to analyse a minimum of 10% of all samples per session in duplicate. We select the samples for duplication before the session begins. The extract is analysed twice independently, not two readings of the same loaded cuvette.

For measurements, we should follow these quality assurance steps:

- Calculate the coefficient of variation for each pair:

$$CV(\%) = \left(\frac{\text{standard deviation}}{\text{mean}} \right) * 100$$

- The acceptance criterion is equal to or lower than 5%. If the CV exceeds 5%, both replicates need to be re-run.
- If a second pair of replicates also fails, flag the sample. Exclude this value from the dataset until the cause is identified. Make a record of the deviation [31].

Calculations and unit conversion

For each sample, these steps must be completed after the spectrophotometer reading is obtained:

- Correct for dilution: we multiply the measured concentration by the dilution factor applied during sample preparation.
- Convert to dry-weight basis:

$$\text{Concentration} = \frac{\text{Corrected value} * \text{exact volume}}{\text{Sample mass}}$$

- We record both the extract volume and dry mass for every sample to ensure the calculation is reproducible.
- Enter the values into the shared project Excel spreadsheet in Teams within 48 hours of the analysis session.

Concentrations outside of the measuring range should not be recorded. In the case of too high a concentration, the sample should be diluted and tested again. If something does not go as planned, like the kit is old, the blank test did not work, or the results are not very similar, the Concentrations results must be noted in the remarks section, and the lead analyst needs to know about the Concentrations results before they are used.

3.7.4 Data handling and management

Sample storage

If not directly used, the sample leaves are stored in brown paper bags and refrigerated for no longer than 24 hours before the laboratory analysis is performed to not affect the outcome of the results.

Data processing

Field and laboratory data are put into the shared project Excel spreadsheet in Teams within 48 hours of each site visit. A second team member will then check all the entries within a week to ensure no mistakes were made by matching the sample IDs to the field sheets.

3.8 Supporting Reference Data

Nutrient sufficiency ranges and deficiency classification

The table below lists the sufficient, marginal and deficient concentration ranges of each of the nine expected nutrients in the leaves. The concentrations are expressed in mg/kg dry matter. The ‘Sufficient range’ column outlines concentrations that support optimal tree health and normal physiological function. The ‘Marginal’ range indicates values at which nutrient stress may start to affect tree performance. The ‘Deficient’ threshold shows values at which visual deficiency symptoms are likely to appear. To use this table, compare the measured concentrations from the performed laboratory tests and compare them with the limits below. These ranges are specific to Oak leaf tissue sampled during the growing season and should not be applied to other species or parts of the tree [5] [32].

Table 14 Nutrient sufficiency ranges and deficiency classification

Element	Sufficient range (mg/kg DM)	Marginal (mg/kg)	Deficient (< mg/kg)	Source
N	18,000–25,000	15,000–18,000	< 15,000	[5]
P	1,200–2,500	800–1,200	< 800	[33]
K	7,000–15,000	5,000–7,000	< 5,000	[5]
Ca	8,000–20,000	5,000–8,000	< 5,000	[32]
Mg	1,500–3,500	1,000–1,500	< 1,000	[5], [32]
Fe	80–200	50–80	< 50	[34]
Mn	30–300	20–30	< 20	[5]
Zn	15–50	10–15	< 10	[34]
Cu	5–15	3–5	< 3	[34]

Nutrient ratio indices

Nutrient ratios help identify imbalances between elements that individual concentrations cannot depict on their own. This table includes four important ratios (N:P, K:Mg, Ca:Mg, Fe:Mn), their ideal ranges for *Quercus robur*, and advice on interpreting values outside these ideal ranges. To calculate a ratio, divide the concentration of the first element by the

concentration of the second element, both in the same units (mg/kg dry matter). Then, compare the result to the "Optimal range" column. If the calculated ratio is outside the optimal range, refer to the "Interpretation of imbalance" column to understand the possible physiological impact. Ratios should be calculated and checked in the master Excel spreadsheet after each monitoring round, along with the individual element comparisons against the sufficiency table above.

Table 15 Nutrient ratio indices

Ratio	Optimal range	Interpretation of imbalance	Source
N:P	10–15	> 15 suggests P limitation; < 10 suggests N limitation	[35]
K:Mg	2–3	> 4 may indicate Mg deficiency induced by high K	[35]
Ca:Mg	3–6	> 10 may indicate Mg deficiency or Ca toxicity	[35]
Fe:Mn	1–4	Imbalance may affect enzyme function and photosynthesis	[35]

3.9 Tree canopy sampling - climbing and safety protocol

This protocol describes the safety procedures for tree canopy sampling using a long extension ladder and pole pruner, conducted in the absence of conventional top-rope climbing access. All personnel must read and understand this protocol before commencing work. In the absence of traditional arboricultural climbing access, ladder-assisted entry with a dedicated rope belay system is a recognised and effective method for working at height in trees [36], [37]. The system complies with EN 1891 (static rope standard) and relevant PPE regulations for working at height.

3.9.1 Scope and applicability

This protocol applies to:

- Tree height: 20–25 metres
- Access method: extension ladder secured to tree
- Sampling method: pole pruner operated from ladder
- Minimum personnel: 2 persons (climber + belayer)
- Weather conditions: dry, wind speed < Beaufort 4

3.9.2 Equipment

All equipment must be inspected before use. Any damaged or expired equipment must be replaced before the operation commences (Arboricultural Association, 2020).

Item	Specification	Status
Full-body harness (x2)	EN 361 certified	☐ Checked
Mammut Smart 2.0 belay device	Compatible ropes: 8.7–10.5 mm	☐ Checked
Mammut Smart HMS carabiner	Triple-lock, EN 362	☐ Checked
HMS Triact Lock carabiner	Triple-lock, EN 362	☐ Checked
Static rope — main	10.5 mm, 60 m, EN 1891 Type A	☐ Checked
Static rope — anchor sling	10.5 mm, 2 m, loop	☐ Checked
Extension ladder	Secured to tree with strap/chain	☐ Checked
Pole pruner	Inspect blade and connections	☐ Checked
Protective gloves (x2 pairs)	Cut-resistant; for handling rope, pole pruner and branch samples	☐ Checked
Insect/tick repellent	DEET-based or equivalent; for exposed skin and clothing	☐ Checked
Tick removal tool / fine-tipped tweezers	For prompt removal of attached ticks	☐ Checked
Sample bags + labels	Sufficient quantity for field session	☐ Checked
First aid kit	Including emergency contact card	☐ Checked

Mobile phone (charged)	Emergency services: 112	☐ Checked
------------------------	-------------------------	----------------------------------

Rope specifications. The main rope must conform to EN 1891 Type A (low-stretch kernmantle rope): diameter 10.5 mm; minimum breaking strength 22 kN (selected rope 32 kN); maximum elongation at 50–150 kg load $\leq 5\%$ (selected rope 3.4%); material polyamide (PA), melting point $> 195^{\circ}\text{C}$. Static (low-stretch) rope is preferred over dynamic rope because it minimises vertical movement under working load, giving greater stability when the operator works hands-free with the pole pruner [38].

3.9.3 Personnel and roles

A minimum of two persons is required at all times during the climbing operation. No single-person operation is permitted [39] [40].

Role	Person	Responsibilities
Climber	Person A	Ascends ladder, operates pole pruner, collects samples
Belayer	Person B	Manages rope via Mammut Smart 2.0, maintains brake hand at all times, monitors climber

3.9.4 Personal protective equipment: gloves and insect/tick safety

Gloves. Cut-resistant protective gloves must be worn by both the climber and the belayer throughout the operation. Gloves protect the hands when handling the running rope, operating and carrying the pole pruner, and collecting and bagging cut branch samples (which may have sharp ends or rough bark). Each person should carry a spare pair in case gloves become wet, torn or contaminated.

Insect and tick safety. Fieldwork in and around trees and undergrowth carries a risk of bites from ticks, which can transmit Lyme disease and other tick-borne illnesses, as well as exposure to wasps, mosquitoes and other biting insects. The following measures apply:

- Wear long sleeves and long trousers; tuck trouser legs into socks and tuck shirts in to reduce exposed skin.

- Apply insect/tick repellent (DEET-based or equivalent) to exposed skin and to clothing before starting.
- Prefer light-coloured clothing so that ticks are easier to spot.
- Carry a tick removal tool or fine-tipped tweezers in the first aid kit.
- Check skin, clothing and equipment for ticks during breaks and again at the end of the field session; pay particular attention to warm areas (behind knees, groin, armpits, hairline).
- If a tick is found attached, remove it promptly by gripping close to the skin and pulling steadily upward; clean the bite site and record the date and location of the bite.
- Monitor the bite site for several weeks; if a spreading rash, fever or flu-like symptoms develop, seek medical advice and mention the tick exposure.
- Be alert to wasp/bee nests in the tree or on the ground; anyone with a known severe allergy must carry their own auto-injector and inform the team before work begins.

3.9.5 Step-by-step field protocol

Step 1 - Site assessment

- Visual inspection of the target tree for structural defects, dead wood, fungi or rot.
- Assessment of ground conditions: stability, slope, root systems and trip hazards.
- Identification of a suitable anchor branch: live, structurally sound, minimum arm thickness (~10 cm diameter), positioned as high as safely reachable from the ladder.
- Assessment of weather: no operation in rain, high wind (Beaufort > 4), lightning risk or frost.
- Identification of an emergency egress route from the ladder.

Step 2 - Ladder setup

- Place the ladder base on firm, level ground using the 4:1 rule; for every 4 m of height, the base is 1 m from the tree [41].
- Secure the top of the ladder to the tree trunk using a strap or rope; do not rely on friction alone [39], [41].
- Both persons inspect and confirm the ladder is stable before any climbing.
- Ensure the ladder extends at least 1 m above the intended working height.

Step 3 - Anchor system setup

- The climber ascends the ladder with the anchor sling (2 m loop of 10.5 mm static rope) and one HMS Triact Lock carabiner.

- At height, the sling is manually placed over the selected anchor branch.
- The HMS Triact Lock carabiner is clipped into the bottom of the sling.
- The 60 m main rope is fed through the carabiner, with both ends hanging to the ground.
- The belayer confirms both rope ends are clear and free-running.

Step 4 - Belay system setup

- The belayer puts on a harness.
- The Mammut Smart 2.0 is attached to the belayer's belay loop using the Mammut Smart HMS carabiner (triple-lock, closed and verified).
- The rope is threaded through the Smart 2.0 according to the manufacturer's instructions.
- The belayer takes position directly below or slightly to the side of the ladder, with a clear line of sight to the climber.
- The belayer confirms readiness: brake hand on rope at all times.

Step 5 - Climber setup

- The climber put on a harness.
- The rope is attached to the climber's belay loop using a figure-eight on a bight or clove hitch.
- Both persons check each other's harness, knots and carabiner locks (buddy check).
- Verbal confirmation before ascent: climber states "Climbing" and belayer responds "Climb on."

Step 6 - Ascent and sampling

- The climber ascends the ladder slowly and steadily, maintaining three points of contact where possible.
- The belayer takes in rope slack continuously as the climber ascends.
- Upon reaching working height, the climber verbally confirms position.
- The climber operates the pole pruner to collect branch samples; during pruner operation both hands are occupied and the belay system carries full body weight.
- Cut samples are lowered to the ground in a sample bag using a hand-line; samples must not be thrown.
- Branch offcuts must be called out ("Below!") before they fall, and the belayer must step clear.

Step 7 - Descent

- The climber verbally signals “Descending.”
- The belayer maintains tension on the rope throughout descent.
- The climber descends the ladder slowly and deliberately.
- Upon reaching the ground, the climber calls “Off rope.”
- The belayer may then release the brake hand.

Step 8 - Pack-down

- The climber removes the anchor sling and carabiner from the branch.
- The ladder is untied and lowered safely; both persons handle this together.
- All equipment is inspected for damage, coiled/packed and returned.
- Any equipment that sustained a load-bearing fall must be retired and not reused.

3.9.6 Emergency procedures

In the event of a fall

- The belayer holds the brake; do not release the rope under any circumstances.
- Call emergency services immediately: 112.
- Do not move the casualty unless there is immediate danger.
- Keep the casualty calm and conscious; maintain verbal contact.
- One person stays with the casualty; the other directs emergency services to the location.

Suspension trauma. If the climber is suspended in the harness for more than a few minutes following a fall, suspension trauma (orthostatic shock) may occur. Emergency services must be contacted immediately. The climber should be encouraged to push against the ladder or tree to relieve harness pressure on the legs where possible [36].

Bibliography

[1]

K. H. Mellert and A. Göttlein, "Comparison of new foliar nutrient thresholds derived from van den Burg's literature compilation with established central European references," Sept. 2012, doi: 10.1007/s10342-012-0615-8. Available: <https://doi.org/10.1007/s10342-012-0615-8>. [Accessed: May 27, 2026]

[2]

N. Government, "Plant nutrients in the soil," 2024. Available: <https://www.dpi.nsw.gov.au/agriculture/soils/soil-testing-and-analysis/plant-nutrients>. [Accessed: May 27, 2026]

[3]

Colourworker, "Colourworker - Spectral Reflectance and Colour", Available: <https://www.colourworker.com/background.php>. [Accessed: May 26, 2026]

[4]

M. Hoenig, "Preparation steps in environmental trace element analysis -facts and traps."

[5]

S. A. Jasmini, B. Nasir, N. Hayati, and N. Edy, "Improvement of soil quality using bokashi composting and NPK fertilizer to increase shallot yield on dry land", Available: https://www.cropj.com/lasmini_12_11_2018_1743_1749.pdf. [Accessed: May 26, 2026]

[6]

"Een Natuurgraf op Huis ter Heide." Available: <https://www.natuurbegravennederland.nl/huis-ter-heide/een-natuurgraf/>. [Accessed: May 07, 2026]

[7]

U. EPA, "MICROWAVE ASSISTED ACID DIGESTION OF SEDIMENTS, SLUDGES, SOILS, AND OILS." Feb. 01, 2007.

[8]

"LCK348 Phosphorus total / Phosphate ortho."

[9]

U. EPA, “MICROWAVE ASSISTED ACID DIGESTION OF SILICEOUS AND ORGANICALLY BASED MATRICES.” Dec. 01, 1996.

[10]

“ISO 11885 Water quality — Determination of selected elements by inductively coupled plasma optical emission spectrometry.” Available:
<https://www.iso.org/standard/36250.html>. [Accessed: May 27, 2026]

[11]

B. Martin and C. C. Hach, “An Introduction to Standards and Quality Control for the Laboratory.”

[12]

J. Eichhorn *et al.*, “Visual Assessment of Crown Condition and Damaging Agents.” ICP, 2020. Available: https://www.icp-forests.net/fileadmin/icp_forests/Dateien/Manual_Versions/2025-2026/ICPF_Manual_part04_2020-4_Crown.pdf. [Accessed: May 07, 2026]

[13]

S. L. R. Ellison and A. Williams, “Quantifying Uncertainty in Analytical Measurement,” 2023, Available:
https://www.eurachem.org/images/stories/Guides/pdf/QUAM2012_P1.pdf. [Accessed: May 27, 2026]

[14]

Cantwell. H, “The Fitness for Purpose of Analytical Methods,” 2025, Available:
https://www.eurachem.org/images/stories/Guides/pdf/MV_guide_3rd_ed_V1_EN.pdf. [Accessed: May 27, 2026]

[15]

R. J. Mitchell *et al.*, “Collapsing foundations: The ecology of the British oak, implications of its decline and mitigation options,” May 2019, doi: 10.1016/j.biocon.2019.03.040. Available: <https://www.sciencedirect.com/science/article/pii/S0006320718317920>. [Accessed: May 07, 2026]

[16]

S. Cheng *et al.*, “Beyond crown volume: UAV-LiDAR-derived foliated crown volume (FCV) enhances coniferous tree attribute estimation,” Apr. 2026, doi:

10.1016/j.compag.2026.111555. Available:
<https://www.sciencedirect.com/science/article/pii/S016816992600150X>. [Accessed: May 26, 2026]

[17]

J. Hubert, R. Worrel, C. J. Nixon, and A. Broome, “Pedunculate oak (POK).” Available:
<https://www.forestresearch.gov.uk/tools-and-resources/tree-species-database/131559-pedunculate-oak-pok-2/>. [Accessed: May 07, 2026]

[18]

“David W. Galbraith.”

[19]

M. J. Ambrose and B. L. Conkling, “Forest Health Monitoring 2006 National Technical Report,” July 2009, Available: https://www.srs.fs.usda.gov/pubs/gtr/gtr_srs117.pdf. [Accessed: May 26, 2026]

[20]

M. Hassan, “Handbook of reference methods for plant analysis”, Available:
https://www.academia.edu/5219993/Handbook_of_reference_methods_for_plant_analysis. [Accessed: May 27, 2026]

[21]

K. A. Roubelakis and W. M. Kliever, “Enzymes of Krebs-Henseleit Cycle in *Vitis vinifera* L: II. Arginosuccinate Synthetase and Lyase,” Sept. 1978, doi: 10.1104/pp.62.3.340. Available: <https://doi.org/10.1104/pp.62.3.340>. [Accessed: May 27, 2026]

[22]

B. C. M. of Agriculture, “Leaf Tissue Analysis and Interpretation | BC Tree Fruit Production Guide”, Available: <https://www.bctfpg.ca/horticulture/fruit-tree-nutrition/assessing-nutrient-requirements-in-fruit-trees-sampling-testing-and-interpretation/leaf-tissue-analysis-and-interpretation/>. [Accessed: May 27, 2026]

[23]

R. Thomas, M. Gaithersburg, M. Dekker, and E. K K E R, “Practical Guide to ICP-MS.”

[24]

R. Lindenmayer, L. Lu, F. Eivazi, and Z. Afrasiabi, “Atomic Spectroscopy-Based Analysis of Heavy Metals in Seaweed Species.” in Applied Sciences. MDPI AG, Apr. 10, 2023. doi: 10.3390/app13084764

[25]

D. Osorio, J. Anderson, M. Garvie, and H. Okamoto, “ColourWorker – Turn Your Phone Into a Scientific Sensor.” Available: <https://www.colourworker.com/about.php>. [Accessed: May 07, 2026]

[26]

“ATOMIC SPECTROSCOPY APPLICATIONS IN THE CONTRACT ENVIRONMENTAL LABORATORY Atomic Spectroscopy Solutions for Environmental Applications > Search entire document.”

[27]

J. Eichhorn *et al.*, “Visual Assessment of Crown Condition and Damaging Agents,” June 2023, Available: https://www.icp-forests.net/fileadmin/icp_forests/Dateien/Manual_Versions/2020-22/ICP_Manual_part04_2020_Crown_version_2020-3_update_06-2023.pdf. [Accessed: May 26, 2026]

[28]

“Method 200.7, Revision 4.4: Determination of Metals and Trace Elements in Water and Wastes by Inductively Coupled Plasma-Atomic Emission Spectrometry METHOD 200.7 DETERMINATION OF METALS AND TRACE ELEMENTS IN WATER AND WASTES BY INDUCTIVELY COUPLED PLASMA-ATOMIC EMISSION SPECTROMETRY Revision 4.4 EMMC Version.”

[29]

Z. Yan, D. Tian, W. Han, Z. Tang, and J. Fang, “An assessment on the uncertainty of the nitrogen to phosphorus ratio as a threshold for nutrient limitation in plants,” Nov. 2017, doi: 10.1093/aob/mcx106. Available: <https://doi.org/10.1093/aob/mcx106>. [Accessed: May 27, 2026]

[30]

E. E. Agency, “Old acidophilous oak woods with *Quercus robur* on sandy plains.” Available: <https://eunis.eea.europa.eu/habitats/10193>

[31]

J. Anderson *et al.*, “Image-based leaf SPAD value and chlorophyll measurement using a mobile phone: enabling accessible and sustainable crop management,” Apr. 2026. doi: <https://doi.org/10.1101/2025.07.08.663527>. Available: <https://www.biorxiv.org/content/10.1101/2025.07.08.663527v2>. [Accessed: May 07, 2026]

[32]

R. J. Porra, “The chequered history of the development and use of simultaneous equations for the accurate determination of chlorophylls a and b,” 2002, doi: 10.1023/A:1020470224740. Available: <https://pubmed.ncbi.nlm.nih.gov/16245116/>. [Accessed: May 27, 2026]

[33]

E. Commission, “Pedunculate oak”, Available: <https://forest.jrc.ec.europa.eu/en/european-atlas/qr-trees/pedunculate-oak/>. [Accessed: May 27, 2026]

[34]

“LCK 238 LATON,” 2017, Available: https://cdn.hach.com/7FYZVWYB/at/bb59cbbqnwh2bctn7t6wb6r/DOC3125394006_Ed4_LCK238.pdf. [Accessed: May 27, 2026]

[35]

“Welcome to the next Euroanalysis 2025 in Barcelona, a forum to share knowledge and experience, a think tank and a place to promote Analytical Chemistry networking.”

[36]

Colourworker, “ColourWorker – User Guide”, Available: <https://www.colourworker.com/userguide.php>. [Accessed: May 26, 2026]

Appendices

Appendix 1 – MP-AES foliar nutrient concentrations

Elemental concentrations from acid-digested leaf samples, expressed on a dry-weight basis. Compare results against the sufficiency ranges in Table 11 of the monitoring plan.

Tree ID	Round	N (mg/kg DM)	P (mg/kg DM)	K (mg/kg DM)	Ca (mg/kg DM)	Mg (mg/kg DM)	Fe (mg/kg DM)	Mn (mg/kg DM)	Zn (mg/kg DM)	Cu (mg/kg DM)
T01	R1									
T01	R2									
T01	R3									
T02	R1									
T02	R2									
T02	R3									
T03	R1									
T03	R2									
T03	R3									
T04	R1									
T04	R2									
T04	R3									
T05	R1									
T05	R2									
T05	R3									
T06	R1									
T06	R2									
T06	R3									
T07	R1									
T07	R2									
T07	R3									

Appendix 2 – PhotoFolia Leaf Greenness Values

LGV readings taken from four crown positions (N, E, S, W) per tree per round. The per-tree mean leaf greenness value is the analytical unit for chlorophyll comparisons.

Tree ID	Round	LGV N	LGV E	LGV S	LGV W	Mean LGV
T01	R1					
T01	R2					
T01	R3					
T02	R1					
T02	R2					
T02	R3					
T03	R1					
T03	R2					
T03	R3					
T04	R1					
T04	R2					
T04	R3					
T05	R1					
T05	R2					
T05	R3					
T06	R1					
T06	R2					
T06	R3					
T07	R1					
T07	R2					
T07	R3					

Appendix 3 – Visual crown assessment scores

Scores recorded during each monitoring round following the ICP Forests crown condition protocol (Section 3.4 of the monitoring plan). Final agreed scores between both observers should be entered here.

Tree ID	Round	Date	Defoliation class (0–4)	Crown transparency (%)	Crown dieback (%)	Epicormic growth (0–2)	Remarks
T01	R1						
T01	R2						
T01	R3						
T02	R1						
T02	R2						
T02	R3						
T03	R1						
T03	R2						
T03	R3						
T04	R1						
T04	R2						
T04	R3						
T05	R1						
T05	R2						
T05	R3						
T06	R1						
T06	R2						
T06	R3						
T07	R1						
T07	R2						
T07	R3						

Appendix 4 – Hach Lange LCK 238 & 348 results (N and P)

Colorimetric cuvette test results for total nitrogen (LCK 238) and reactive phosphate (LCK 348). Corrected values are back-calculated to mg/kg DM using the dilution factor, extract volume and sample dry mass (see Section 3.6.5 of the monitoring plan).

Tree ID	Round	Sample ID	N – LCK 238 (mg/L measured)	Dilution factor (N)	N corrected (mg/kg DM)	P – LCK 348 (mg/L measured)	Dilution factor (P)	P corrected (mg/kg DM)
T01	R1							
T01	R2							
T01	R3							
T02	R1							
T02	R2							
T02	R3							
T03	R1							
T03	R2							
T03	R3							
T04	R1							
T04	R2							
T04	R3							
T05	R1							
T05	R2							
T05	R3							
T06	R1							
T06	R2							
T06	R3							
T07	R1							
T07	R2							
T07	R3							

Appendix 5 – Calculated nutrient ratios

Ratios calculated from ICP-OES results in Table A1. Optimal ranges for *Quercus robur* are shown in the column headers. Values outside the optimal range should be flagged in the remarks column and cross-checked against Table 12 of the monitoring plan.

Tree ID	Round	N:P (optimal 10–15)	K:Mg (optimal 2–3)	Ca:Mg (optimal 3–6)	Fe:Mn (optimal 1–4)	Remarks
T01	R1					
T01	R2					
T01	R3					
T02	R1					
T02	R2					
T02	R3					
T03	R1					
T03	R2					
T03	R3					
T04	R1					
T04	R2					
T04	R3					
T05	R1					
T05	R2					
T05	R3					
T06	R1					
T06	R2					
T06	R3					
T07	R1					
T07	R2					
T07	R3					